

A2 Notes P-5 Advanced Probability

I. Probability of A AND B: $P(A \text{ and } B)$

To find the probability of two events occurring together, you have to decide whether one event occurring affects the other event.

If one event DOES affect how the second event can occur, we call them dependent events.

If it does NOT affect the second event, we call them independent events.

Consider these scenarios:

1. Event 1: Roll a die Event 2: Flip a coin

These are independent events.

2. Event 1: Selecting a shirt from your closet to pack.
Event 2: Selecting another shirt from your closet to pack.

These are dependent events.

If event A and event B are INDEPENDENT, we find the probability of them BOTH happening by:

$$P(A \text{ and } B) = \frac{P(A) \cdot P(B)}{\quad \cdot \quad}$$

Examples:

1. At a picnic, there is a drink tub and a snack tub. There are 10 diet drinks and 5 regular drinks. There are 8 bags of chips and 12 bags of pretzels. If you choose a drink and a snack, what is the probability of... Need TOTALS! Drinks=15 Snacks=20

a. $P(\text{diet and pretzels})$

$$\begin{aligned} &= P(\text{diet}) \cdot P(\text{pretzel}) \\ &= \frac{10}{15} \cdot \frac{12}{20} = \frac{4}{10} = \boxed{\frac{2}{5}} \end{aligned}$$

$$\frac{10}{20} = \frac{(10)(1)}{(10)(2)}$$

You! b. $P(\text{regular and chips})$

$$\frac{5}{15} \cdot \frac{8}{20} = \frac{8}{60} = \boxed{\frac{2}{15}}$$

2. There is a spinner with 14 equal areas, numbered 1 through 14. If the spinner is spun one time, what is the probability that the result is a multiple of 3 and a multiple of 4?

mult 3: 3, 6, 9, 12
mult 4: 4, 8, 12

$$P(\text{mult 3 and mult 4}) = \frac{1}{14}$$

at the same time

II. Probability of A OR B: $P(A \text{ or } B)$

To find the probability of either event A OR event B occurring, you have to decide whether the events are mutually exclusive. If events are mutually exclusive, they cannot happen at the same time.

Consider these scenarios:

1. You roll a standard die (with #1-6) one time. Can you roll a 2 AND a 3? No

Therefore, these (circle one) are / aren't mutually exclusive events.

2. You roll the die again. Can you roll an even number AND a multiple of 3? Yes

Therefore, these (circle one) are / aren't mutually exclusive events.

If event A and event B are **MUTUALLY EXCLUSIVE**:

$$P(A \text{ or } B) = P(A) + P(B)$$

If event A and event B are **NOT MUTUALLY EXCLUSIVE**:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

overlap!

Examples:

1. About 30% of the U.S. population is under 20 years old and roughly 17% of the population over the age of 60. What is the probability that a person at random is under 20 or over 60?

case 1 <20 yrs old case 2 over 60 =
→ mutually exclusive

$$P(<20 \text{ or } >60) = P(<20) + P(>60)$$

$$= .30 + .17 = .47$$

2. In a certain Algebra 2 class of 24 students, 16 of them play basketball and 6 of them play football. There are 6 students who play neither sport. What is the probability that a student chosen randomly from the class plays both basketball and football?

① Make a chart

	F	No F	TOTAL
BB.	4	12	16
No BB	2	4	8 (24-16)
TOTAL	6	18	24

② Fill in what you know

③ Solve for empty boxes

④ Answer! $\frac{4}{24} = \frac{1}{6}$

TOTAL * STUDENTS

24-6

3. There is a spinner with 14 equal areas, numbered 1 through 14. If the spinner is spun one time, what is the probability that the result is a multiple of 2 or a multiple of 6?

mult of 2: 2, 4, 6, 8, 10, 12, 14

mult of 6: 6, 12
overlap!

$$= \frac{7}{14} = \frac{1}{2}$$

4. In a class of 27 sophomores and juniors, 13 are sophomores and 18 students have an A. There are 5 students who are juniors and do not have an A. What is the probability that: (Hint: make a chart!)

a. a student does not have an A?

	A	not A	TOTAL
soph	9	4	13
junior	9	5	14
TOTAL	18	9	27

$$P(\text{not A}) = \frac{9}{27} = \frac{1}{3}$$

b. a student has an A given that the student is a sophomore?

our total is total sophomores, not total students

$$\frac{9}{13}$$

5. A bag contains 12 blue marbles, 7 purple marbles, and 9 yellow marbles. If two marbles are drawn out of the bag, what is the probability that both are blue:

Need total = 28

a. exactly.

(Answer as a fraction)

$$\frac{12}{28} \times \frac{11}{27} = \frac{11}{63}$$

b. to the nearest 1000th.

$$\frac{11}{63} \approx .174603 \dots$$

$$\boxed{.175}$$